

people would use it, and so more residents of that village would be able to get jobs farther away from their homes, or more farmers would be able to sell their produce at higher prices in the city. But looking at congestion is a terrible way to measure the need for more roads, largely because using roads is generally free. If you give away something that costs money to provide, you will almost always get more demand than you can ever possibly meet. And if driving is too cheap relative to other ways of getting around, there will always be more demand to drive. That is why congestion is a problem in almost every big city on earth.

And when we look at cities in rich countries, the reality is that overall, we have almost certainly oversupplied the amount of road needed already. To take another American example, look at Louisville, the biggest city in Kentucky. There is a bridge there, called the John F. Kennedy Memorial Bridge, that crosses the Ohio River, on the I-65 Interstate highway, connecting Louisville to Jeffersonville, a smaller city in Indiana. In 2010, around 120,000 vehicles crossed the bridge daily, which had six lanes of traffic (three in each direction), and routinely got completely clogged up with drivers using it. So the highway engineers at the Kentucky and Indiana departments of transport decided that they needed to widen it. As reported by the City Observatory, a think tank based in Portland, Oregon, together the departments of transport put up \$1 billion to build a new, second bridge (this one named for Abraham Lincoln), adding six more lanes. In 2017, when the bridge opened, to pay back the cost, they introduced a toll to cross both bridges—\$2 for each crossing, with discounts for regular commuters.

And guess what happened? The number of drivers using the bridges fell, dramatically, to around 60,000 cars per day. It has never risen back up to its previous level. What this meant was that the extra lanes proved completely pointless. As the City Observatory reported, "the two states spent a billion dollars doubling the size of I-65, only to have half as many people use the bridge. That money was wasted." What happened to the extra cars? Much as with the collapse of the West Side Elevated Highway, it is almost impossible to know. But the most plausible explanation is that most of them switched to using the Second Street Bridge, another bridge across the river just a few hundred yards away, which had no toll. That adds several minutes to the typical commute, and more at rush hour, but it saves drivers a few bucks. The state departments of

transport thought they were helping drivers by speeding up their commutes, but the drivers actually didn't care. When charged even a modest fee, traffic dropped away. There was already plenty of road to begin with.

There are plenty of other cases that show this is not a one-off. In the UK there is only one major toll road in the whole country, the M6 Toll, which bypasses Birmingham. This motorway was first proposed in the 1980s, when the original M6 motorway was already handling more than twice the traffic it was constructed for. In 2003, the M6 Toll opened to great fanfare, having finally been built by a private consortium, with the tolls intended to cover the cost. In a touch almost perfectly designed to appeal to tabloid newspapers, the construction involved pulping 2.5 million Mills & Boon novels (a brand of racy romance fiction sold in Britain) into the tarmac to aid with its water absorbability. It was expected to carry about 75,000 vehicles per day. In fact, even though traffic in the UK has increased consistently over the past twenty years since it opened, it has never got much above a level of 50,000 vehicles per day. During the great financial crisis of 2008, usage dropped to just 30,000 cars per day.

Studies of toll roads show that they almost always underperform the anticipated usage. America now has plenty of toll roads—mostly outside of urban areas. In the past thirty years, the revenue generated by the federal gas tax, which funded the original Interstate system, has declined dramatically in real terms, as politicians have refused to raise it even once in line with inflation. And so states have tapped into private money to build new highways. To fund them, the developers have to issue bonds—debt—which means that they have to reassure the investors that the road will generate enough revenue. Between 1994 and 2006, public-private partnerships spent \$21 billion on toll roads in America. But according to Standard & Poor's, a debt rating agency, traffic is invariably overestimated by around 30 percent on average. "Optimism bias is a consistent trend in toll-road traffic forecasting," the firm reported in 2002. More recent studies agree. In 2014, the journal *Transportation* reported that "With rare exception, actual toll road traffic in many countries has failed to reproduce forecast traffic levels."

Traffic engineers cannot resist trying to fix congestion with one more road. The reason is exactly why you should not put engineers in charge of social problems, like how to get people around. They tend to want to simplify